Final Report

Introduction to Machine Learning Safety

CARLA Autonomous Driving Perception System

Student: Vivek Chitradurga Mallikarjun

Matriculation number: 261591

Date: 23 August 2026

Otto-von-Guericke University Magdeburg

Contents

Code Repository Submission

All evidence in this report was produced by the code in the publicly accessible repository `https://github.com/vivekcm143/ML_Safety_SoSe26`. The `README.md` in the repository root explains how to reproduce each number, and the reproduction map in Appendix A links every table and figure of this report to the script or notebook that produced it.

The traceability chain used throughout is Evidence ⇒ V ⇒ SC ⇒ LS ⇒ UCA ⇒ H ⇒ L, with the ID conventions of the course template (L = loss, H = hazard, UCA = unsafe control action, LS = causal loss scenario, SC = safety constraint, V = verification).

1. System Description

Provided system description

Reproduced verbatim from the exercise specification.

1. **Camera:** A single forward-facing RGB camera, rigidly mounted behind the windshield, provides images at 10 Hz. There are *no other perception sensors* (no radar, no LiDAR, no ultrasonic).
2. **Perception models:** Three separate binary classifiers each receive the camera frame and output one binary prediction: *pedestrian present*, *vehicle present*, or *traffic light present*. Each model was trained exclusively on sunny daytime data collected in the CARLA simulator.
3. **Distance oracle:** A separate model estimates the distance to detected pedestrians and vehicles. You may assume this model is perfect for the sake of this course.
4. **Planning module:** The rule-based planner consumes the three perception outputs together with the current vehicle speed and steering angle. Its primary safety-relevant decisions are whether to continue driving, request deceleration, or issue an emergency brake command. Emergency braking is triggered when the pedestrian detector reports a pedestrian within a critical distance, when the vehicle detector indicates a road user ahead within a speed-dependent threshold, or when the traffic-light detector indicates that stopping is required at an approaching intersection. The planner does not have access to raw camera images and relies entirely on the perception models' outputs.
5. **Vehicle actuators:** Throttle, brake, and steering commands are executed by drive-by-wire actuators. Emergency braking applies maximum deceleration within the physical limits of the vehicle.
6. **Human safety operator:** A trained operator sits in the driver seat and monitors the road. The operator can override the autopilot at any time by pressing the brake pedal or turning the steering wheel. The operator receives a visual dashboard showing the current perception output and system status, but no auditory alerts are provided. During testing, operators work 4-hour shifts.
7. **Operating environment:** The vehicle is tested on public urban roads at speeds up to 50 km/h. The intended conditions are daytime, dry weather, and mapped intersections. However, weather and lighting conditions can change during a test drive (e.g. sudden cloud cover, low sun angle, rain onset).

Key design limitations (by construction)

- No sensor redundancy — a single camera is the only perception input.
- No other safety mechanisms are deployed yet.
- The human operator is the only fallback if the automation fails.
- The operator may exhibit a non-zero probability of delayed or missed intervention, especially under prolonged monitoring.

The traffic-light detector reports only the *presence* of a traffic light, not its state (red / green). The planner therefore cannot, on its own, decide whether to stop at a signalized intersection. This is recorded as known design limitation DL-1 in Section 5; no model metric in this report can close it.

Architecture & Training

All three detectors share *one* architecture: an ImageNet-pretrained **ResNet-18** whose 1000-way classification head is replaced by a single `Linear(512, 1)` unit, so each network emits one logit that is turned into a probability with a sigmoid. The three networks are trained *independently* on the same images with three different label columns. Separate single-task models were chosen deliberately for safety reasons: a fault in one detector cannot propagate through shared weights into the other two, each model can be re-qualified in isolation, and a per-class operating threshold can be tuned without affecting the remaining classes.

Item	Setting
Backbone	ResNet-18, ImageNet-pretrained, all layers fine-tuned
Head	`Linear(512, 1)` $\rightarrow$ sigmoid
Loss	`BCEWithLogitsLoss` (binary cross-entropy on logits)
Optimiser	Adam, lr $= 10^{-3}$ (baseline detectors), 10^{-4} (Ex.-5 model)
Epochs / batch size	5 / 32
Pre-processing	`Resize(224,224)` $\rightarrow$ `ToTensor` $\rightarrow$ `Normalize`(μ, σ = ImageNet)
Augmentation	none (baseline detectors); random horizontal flip (Ex.-5 model)
Hardware	NVIDIA GeForce RTX 3050 Laptop GPU (CUDA), PyTorch 2.11
Train split	7 200 frames
Validation split	3 600 frames
Test split (in-ODD)	3 600 frames
Shift splits	`test-fog`, `test-night`, `test-town-01`: 3 600 frames each

Class balance. The training split is imbalanced in opposite directions for the three tasks: pedestrian 1 718 positive / 5 482 negative (23.9 % positive), vehicle 5 458 / 1 742 (75.8 %), traffic light 5 276 / 1 924 (73.3 %). The pedestrian task is therefore both the rarest *and* the most safety-critical, which is the direct cause of the recall failure reported in V-1.

Convergence. Training loss fell monotonically for all three models, but only the traffic-light detector also converged on validation (0.113 $\rightarrow$ 0.077); the pedestrian detector diverged on validation (0.569 $\rightarrow$ 0.714) while its training loss kept falling (0.513 $\rightarrow$ 0.269), i.e. it overfits after about three epochs (Table 1).

Second pedestrian model. A separately trained pedestrian detector (`best_model.pth`, Adam 10^{-4}, horizontal-flip augmentation, best-validation checkpointing, *no* ImageNet normalisation, best validation accuracy 0.7367) is used for the temperature/confidence study of V-3 and for all Grad-CAM overlays. It is labelled explicitly wherever it appears, because mixing it with the baseline detectors would break the traceability of the numbers.

2. Operational Design Domain

ODD specification

Dimension	Operating conditions	Non-operating conditions	Detection
Weather	Clear and dry; light overcast	Fog, rain, snow, standing water, spray	k-NN feature monitor (V-4); operator report
Lighting	Daytime, sun elevation $> 20°$, diffuse illumination	Night, dawn/dusk, direct low-sun glare, tunnels	k-NN feature monitor (V-4)

Dimension	Operating conditions	Non-operating conditions	Detection
Camera	Single forward RGB, 10 Hz, clean lens, fixed intrinsics, frames resized to 224×224	Soiled, wet, occluded or defocused lens; dropped frames; changed intrinsics	**none** — no image-quality monitor
Scene type	Mapped urban streets of the training town; signalised intersections	Unmapped towns, motorways, rural roads, construction zones	k-NN feature monitor (V-4), partially
Road users	Pedestrians and passenger cars/vans as sampled by CARLA	Cyclists, motorcycles, animals, prams, wheelchairs	**none** — no class-novelty monitor
Speed range	0–50 km/h	> 50 km/h	Vehicle speed signal (deterministic)
Traffic-light state	— (detector reports presence only)	Any situation requiring red/green discrimination	**none** — design limitation DL-1
Input pipeline	Inference pre-processing bit-identical to training pre-processing	Any deviation (e.g. missing normalisation)	Start-up equivalence test (SC-9), *not implemented*

ODD Gap Analysis

The training split contains *only* clear-weather, daytime frames from a single CARLA town. The following declared ODD dimensions are therefore not covered by the training data at all, and **no performance claim of this report extends to them**: dawn/dusk and low-sun glare (absent from every split); night and fog (present only as dedicated shift splits, never in training); rain, snow and spray (present in no split); any town other than the single training town; cyclists, motorcycles and other vulnerable road users; and any form of camera degradation (soiled, wet or defocused optics).

The measured cost of these gaps is substantial. On the Exercise-5 pedestrian detector accuracy falls from 0.7967 on the in-ODD test split to 0.7517 on the unseen town (−4.5 pp), 0.7636 under fog (−3.3 pp) and 0.7106 at night (−8.6 pp). The different-town shift is *structural* (road layout, buildings, vegetation) under nominal weather, whereas fog and night are *appearance* shifts. The ODD is worded above so that the unseen town is unambiguously **outside** it: the models demonstrably lose accuracy there, so town novelty must be flagged by the monitor rather than silently handled.

ODD Coverage

Coverage is measured with *k-projection coverage*: the ODD is discretised into the seven dimensions above that have a finite value set, and the metric reports the fraction of all k-way combinations of dimension values — over every choice of k dimensions — that occurs in at least one test frame. It is the right metric here because a test suite can look large while exercising each condition only in isolation; the drop of coverage as k grows exposes exactly those missing *interactions*. The full grid has 432 cells, of which the four available test splits realise 96.

Evidence (k-projection coverage), `odd_coverage/k_projection_coverage.py`:

k	Coverage	Uncovered examples
1	$15/17 = 0.882$	rain; dawn/dusk lighting
2	$93/123 = 0.756$	(fog, night); (fog, unseen town); (night, unseen town)
3	$309/491 = 0.629$	(fog, night, unseen town); (rain, ·, ·) for all pairs

Interpretation. Coverage decreases monotonically with k ($0.88 \to 0.76 \to 0.63$), the signature of a test suite built from *one-factor-at-a-time* splits: every non-nominal condition is sampled on its own, but no frame combines two of them — no foggy night, no night in the unseen town, no rain at all. Since combined degradations are exactly where a camera-only stack is weakest, the 37 % of 3-way combinations left uncovered is the most serious limitation of the evidence base. Every model-level verdict in Section 4 is therefore valid only over the covered region; the remainder is carried forward as residual risk RR-6.

ODD Violation Response

Detection of an ODD violation is verified in V-4; the required *response* (speed limitation, minimal-risk manoeuvre, operator takeover request) is a system-level control and is argued in V-5 through constraints SC-6 and SC-7.

3. Safety Analysis (STPA)

Control structure

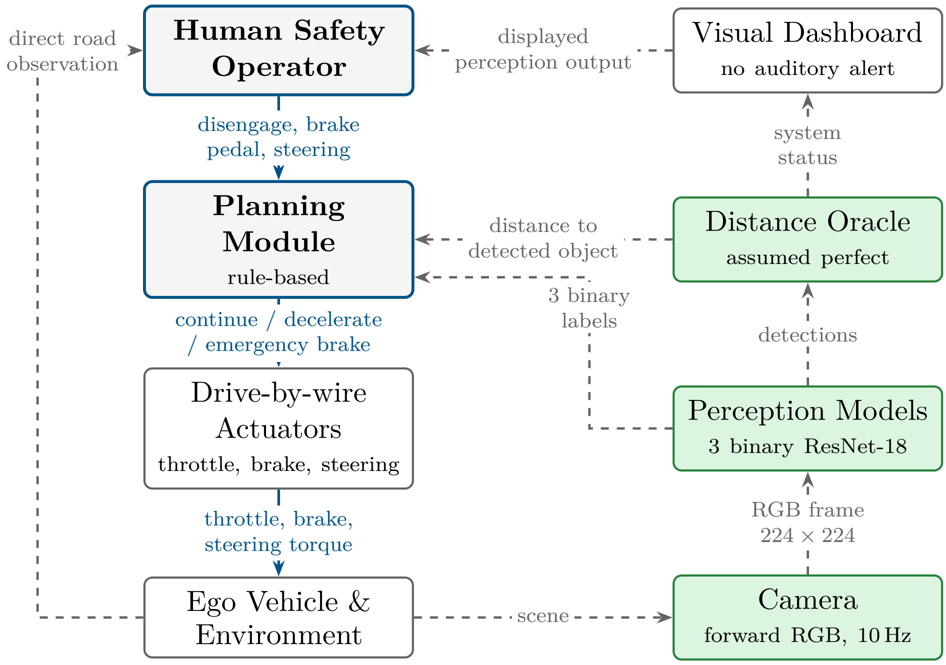


Figure 1: Control structure of the CARLA system. Solid blue arrows are control actions, dashed grey arrows are feedback. The operator is the higher-level controller and its only feedback channels are the visual dashboard and direct observation of the road — there is no auditory alert and no independent sensor.

Losses

ID	Loss	Why unacceptable
L-1	Death or serious injury of a pedestrian or other vulnerable road user	Irreversible harm to an uninvolved party who has not consented to the risk; the dominant societal acceptance criterion for AV testing.
L-2	Death or serious injury of vehicle occupants (ego or third-party)	Irreversible harm; also arises from unnecessary hard braking, not only from collisions.

ID	Loss	Why unacceptable
L-3	Damage to the ego vehicle, third-party property or infrastructure	Reversible but costly; a leading indicator of the mechanisms that cause L-1/L-2.
L-4	Loss of regulatory permission and public trust in the test programme	A single publicised incident terminates the programme, so it must be treated as a first-class loss rather than a consequence.

Hazards

ID	Hazard	Loss(es)
H-1	The ego vehicle approaches a pedestrian in its path without decelerating.	L-1, L-4
H-2	The ego vehicle approaches a stationary or slower vehicle ahead within its stopping distance without decelerating.	L-2, L-3, L-4
H-3	The ego vehicle enters a signalised intersection while it is required to stop.	L-1, L-2, L-3, L-4
H-4	The ego vehicle applies emergency braking when no obstacle is present.	L-2, L-3, L-4
H-5	The ego vehicle continues in automated mode while its camera input lies outside the ODD, so that perception output is untrustworthy and no operator takeover occurs.	L-1, L-2, L-3, L-4

Unsafe Control Actions

ID	Controller	Control action	UCA type	Hazard(s)	Context in which unsafe
UCA-1	Planner	Emergency brake	Not provided	H-1	A pedestrian is within the critical distance in the ego path.
UCA-2	Planner	Emergency brake	Not provided	H-2	A vehicle is ahead within the speed-dependent braking threshold.
UCA-3	Planner	Decelerate / stop at intersection	Not provided	H-3	The vehicle approaches a signalised intersection at which stopping is required.
UCA-4	Planner	Emergency brake	Provided when not needed	H-4	No road user is within the critical distance and a following vehicle is close behind.
UCA-5	Planner	Continue at set speed	Provided	H-5	The camera input is outside the ODD (fog, night, unseen town, soiled lens).
UCA-6	Planner	Emergency brake	Provided too late	H-1, H-2	The obstacle is only reported after the braking distance at the current speed has already been consumed.
UCA-7	Operator	Manual override (brake / steer)	Not provided	H-1, H-2, H-3	The automation has failed to act and a collision is imminent.

ID	Controller	Control action	UCA type	Hazard(s)	Context in which unsafe
UCA-8	Operator	Manual override	Provided too late	H-1, H-2	Late in a 4-hour monitoring shift, with only a visual dashboard and no auditory alert.

Causal Loss Scenarios

LS-ID	For UCA	Causal scenario (why the UCA occurs)
LS-1	UCA-1, UCA-6	The pedestrian detector emits a false negative on a nominal in-ODD daytime frame. Measured recall is 0.191, i.e. 571 of 706 pedestrian-positive test frames produce no detection, so the planner registers "no pedestrian" and never arms the brake.
LS-2	UCA-2	The vehicle detector misses a vehicle ahead (recall 0.800; 540 of 2 700 positive frames missed), so no deceleration request is generated.
LS-3	UCA-3	The traffic-light detector reports only presence, not state. Even a perfect detector leaves the planner unable to decide whether stopping is required.
LS-4	UCA-1, UCA-2	The camera feed is adversarially perturbed — either digitally (ℓ_∞-bounded noise on the frame) or physically (a small patch in the scene). A 10×10 pixel patch is already sufficient to flip the pedestrian output on a compromised model (ASR = 1.00, V-2).
LS-5	UCA-1, UCA-4	Overconfident misclassification: the pedestrian classifier emits a false negative while reporting high confidence (mean predicted-class confidence 0.959 at accuracy 0.804; ECE = 0.156). The planner records "no pedestrian, reliable" and no low-confidence fallback is triggered.
LS-6	UCA-5, UCA-1	Undetected out-of-ODD input: fog, night or unseen-town frames are processed as if in-ODD. Confidence cannot be used as the alarm because the pedestrian model becomes *more* confident on OOD inputs (0.945 in-ODD $\rightarrow$ 0.976 under fog and at night).
LS-7	UCA-7, UCA-8	Operator vigilance decrement: a 4-hour monitoring shift with a purely visual dashboard and no auditory alert delays or suppresses takeover exactly when the automation has already failed.
LS-8	UCA-1, UCA-5	Training/serving pre-processing skew: an inference pipeline that omits the ImageNet normalisation used at training collapses the detectors to a constant negative output (observed: vehicle detector 0 positive predictions on 3 600 frames, accuracy 0.250). Total loss of detection is reported as ordinary, high-confidence "nothing there".
LS-9	UCA-1	Supply-chain data poisoning: 10 % of pedestrian-positive training frames carry a trigger with a flipped label. Clean-set behaviour remains plausible (recall 0.381) while the trigger achieves ASR = 1.00, so acceptance testing on clean data cannot reveal the backdoor.

Safety Constraints

SC-ID	Controls	Safety constraint (requirement)	Level	Verified by
SC-1	LS-1	Pedestrian recall on the in-ODD test set shall be ≥ 0.90.	Model	V-1
SC-2	LS-2	Vehicle recall and traffic-light recall on the in-ODD test set shall each be ≥ 0.85.	Model	V-1
SC-3	LS-4, LS-9	Recall drop under FGSM at $\varepsilon = 0.05$ shall be < 10 pp for every detector, and no input-space trigger shall achieve an attack success rate > 1 %.	Model	V-2

SC-ID	**Controls**	**Safety constraint (requirement)**	**Level**	**Verified by**
SC-4	LS-5	ECE on the in-ODD test set shall be ≤ 0.05 after calibration, and the planner shall brake at the cost-optimal threshold $\tau^* = 1/101$ rather than 0.5.	Model	V-3
SC-5	LS-6	An OOD monitor shall separate in-ODD from out-of-ODD input with AUROC ≥ 0.90, for all three detectors.	Model	V-4
SC-6	LS-6, LS-1	On a detected ODD violation, or when the predicted probability lies in the ambiguous band around $\theta = 0.6$, the system shall limit speed to $\leq 15\,\mathrm{km/h}$ and request operator takeover; if takeover does not occur within the takeover budget it shall execute a minimal-risk stop.	System	V-5
SC-7	LS-7	The takeover request shall be signalled on an auditory *and* a visual channel; continuous monitoring shifts shall be limited and takeover latency shall be measured per operator.	System	V-5
SC-8	LS-3	The planner shall not derive stop decisions at signalised intersections from the presence-only traffic-light detector; a state-aware or map-based input is required.	System	V-5
SC-9	LS-8	The deployed inference pre-processing shall be bit-identical to the training pre-processing, enforced by an automated equivalence test at every start-up.	System	V-5

4. Safety Case

Claim: *The CARLA autonomous driving system operates safely within its Operational Design Domain as specified.*

Verdict: the claim is not supported by the evidence in this report.

Argument strategy. The claim holds if every hazard in Section 3 is adequately controlled by the safety constraints derived from it, so the claim is argued by verifying each constraint below: **model-level** constraints from the experimental evidence produced across the exercises, **system-level** constraints from a design argument. Constraints that cannot be met are carried to Section 5. Every model-level verdict is valid only over the ODD region covered by the test suite (Section 2: 0.63 at $k = 3$).

Verification	Verifies SC(s)	Loss scenario(s)	Hazard(s)	Verdict
V-1: In-distribution detection performance	SC-1, SC-2	LS-1, LS-2	H-1, H-2, H-3	× **Not met**
V-2: Robustness to perturbations	SC-3	LS-4, LS-9	H-1, H-2	× **Not met**
V-3: Calibrated uncertainty	SC-4	LS-5	H-1, H-4	× **Not met**
V-4: Out-of-distribution detection	SC-5	LS-6	H-5, H-1	≈ **Partial**
V-5: Safe system fallback	SC-6–SC-9	LS-3, LS-6, LS-7, LS-8	H-1–H-5	× **Not met**

V-1: In-Distribution Accuracy

Verifies: SC-1, SC-2 **Hazard(s):** H-1, H-2, H-3 **Loss scenario(s):** LS-1, LS-2
Constraint: The model achieves sufficient recall when operating within the ODD.
Metric: Per-class recall on the in-distribution test set (3 600 frames, decision threshold 0.5).

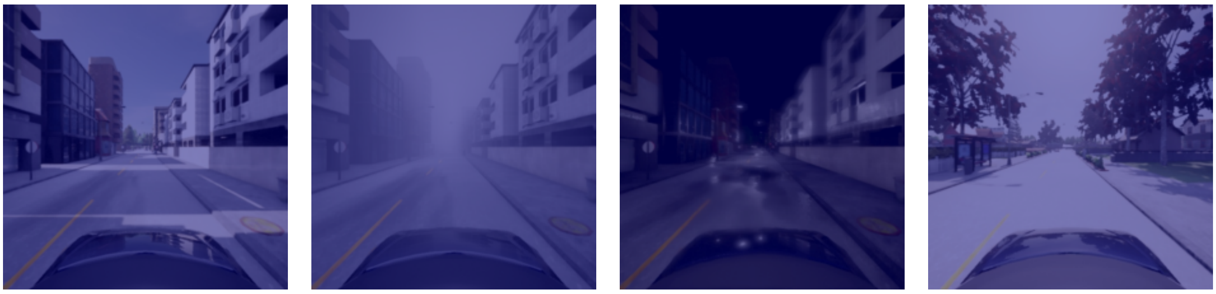
Evidence (In-Distribution Detection Performance)

Model	Recall	Precision	F1	Missed positives	Threshold	Met?
Pedestrian	0.191	0.572	0.287	571 / 706	≥ 0.90	×
Traffic light	0.991	0.916	0.952	24 / 2 584	≥ 0.85	✓
Vehicle	0.800	0.986	0.883	540 / 2 700	≥ 0.85	×

Overall verdict: × **Not met**

The pedestrian detector — the one guarding the most severe loss L-1 — misses 81 % of the frames that contain a pedestrian, yet its accuracy (0.813) still looks acceptable because 80.4 % of test frames contain no pedestrian. Accuracy is dominated by the negative class and is therefore not a valid safety metric here, which confirms that **recall, not accuracy, is the metric that matters**. The vehicle detector is similarly mis-tuned in the unsafe direction (precision 0.986, recall 0.800: it avoids nuisance braking at the cost of 540 missed vehicles), while the traffic-light detector trades false alarms (H-4) for missed stops (H-3) at only 0.916 precision. Both operating points must be re-thresholded (see V-3).

Supporting Evidence: Explanation Quality Grad-CAM (activations of `layer4` weighted by their gradients w.r.t. the single output logit) was applied to the Exercise-5 pedestrian detector. It was chosen because it needs no architectural change, is faithful to the last convolutional stage where spatial evidence is still present, and yields maps directly comparable across conditions.



(a) In-ODD (b) Fog (c) Night (d) New town

Figure 2: Grad-CAM overlays for the pedestrian detector. Panel (a) is inside the ODD; (b)–(d) are outside it. Warm colours indicate high attribution, so a uniform blue tint means the map is essentially flat.

Interpretation. The overlays support LS-1 and LS-6 directly. In-ODD (Fig. 2a) the map is either almost flat or — as in the correctly classified frames of Fig. 4 — concentrated on a building facade in the upper-left corner, never on a road user: the model reaches correct decisions using background structure specific to the training town, the textbook symptom of a spurious correlation and the reason performance transfers so poorly to `test-town-01`. On misclassified frames the map collapses into one large blob centred on a *vehicle* in the middle of the road (Fig. 4), so the pedestrian decision is being driven by vehicle-shaped structure. Under fog, night and the unseen town (Figs. 2b to 2d) attribution becomes near-uniform — no evidence is localised anywhere — yet the model still returns a confident label. Flat attribution combined with rising confidence (V-4) is precisely the mechanism of silent failure.

V-2: Robustness to Input Perturbations

Verifies: SC-3 **Hazard(s):** H-1, H-2 **Loss scenario(s):** LS-4, LS-9
Constraint: Recall does not degrade substantially under small adversarial perturbations, limiting the impact of adversarial manipulation of the camera feed.
Metric: Recall drop under FGSM at $\varepsilon = 0.05$ (clean $\rightarrow$ adversarial).

Evidence (Adversarial Robustness)

Model	Recall (clean)	Recall (FGSM, $\varepsilon = 0.05$)	Drop	Met?
Pedestrian	0.191	not measured	—	× **Not verified**
Traffic light	0.991	not measured	—	× **Not verified**
Vehicle	0.800	not measured	—	× **Not verified**

Substitute evidence for the same constraint (input-space perturbation attacks):

Perturbation (pedestrian detector)	Effect	SC-3 limit
10×10 px red trigger, $p = 10\,\%$ poison	ASR = 1.000, clean recall 0.381	ASR ≤ 0.01
Fog	accuracy $0.797 \to 0.764$ (-3.3 pp)	—
Night	accuracy $0.797 \to 0.711$ (-8.6 pp)	—
Unseen town	accuracy $0.797 \to 0.752$ (-4.5 pp)	—

Overall verdict: × Not met

The FGSM measurement specified by the metric was **not carried out**, so the $\varepsilon = 0.05$ recall drop is unknown and SC-3 is formally unverified for the digital-attack case. An executable harness (`adversarial/fgsm_robustness.py`, $\varepsilon \in \{0.01, 0.05, 0.1\}$, budget applied in raw pixel space with the ImageNet normalisation folded into the model) is provided in the repository so the gap can be closed without re-training; this is the first item of future work.

The second half of SC-3 is, however, decisively *violated*. Poisoning 10 % of pedestrian-positive training frames with a 10×10 pixel red square and flipping their label yields an attack success rate of 1.000: *every* triggered pedestrian-positive test frame is classified as "no pedestrian". The backdoored model retains a clean recall of 0.381 — higher than the honest baseline's 0.191 — so no acceptance test on clean data would raise a flag. That patch is about 0.2 % of a 224×224 frame and could be realised physically as a sticker on street furniture, which makes LS-4 and LS-9 credible rather than theoretical. Together with the natural-corruption degradations the available evidence points unambiguously to non-robustness, so the verdict is recorded as *not met* rather than merely unverified.

V-3: Calibrated Uncertainty

Verifies: SC-4 **Hazard(s):** H-1, H-4 **Loss scenario(s):** LS-5
Constraint: The models provide well-calibrated confidence, so that low-confidence predictions can trigger fallback behaviour rather than silently overcommitting.
Metric: Expected Calibration Error (ECE, 10 equal-width bins) on the in-distribution test set; threshold ≤ 0.05. Temperature T chosen by grid search over 0.5–3.0 in steps of 0.1, on validation NLL.

Evidence (Uncertainty) — ECE of the predicted-class confidence, T from validation NLL:

Model	ECE (single)	T^*	ECE (scaled)	Mean conf.	Accuracy	Met?
Pedestrian	0.156	2.9	0.056	0.959	0.804	×
Traffic light	0.597	3.0	0.394	0.902	0.305	×
Vehicle	0.736	3.0	0.557	0.986	0.250	×

An independent cross-check run on the normalised inference pipeline, scoring the positive-class probability instead, reaches the same conclusion for all three models (Table 2).

Overall verdict: × **Not met**

All three models are **overconfident** in both runs: the signed calibration gap (mean confidence minus accuracy) is positive for every model, and every ECE value exceeds the 0.05 threshold. Temperature scaling helps but does not close the gap; the best case is the pedestrian detector at 0.056, still above the limit. The verdict is therefore invariant to which run is taken as authoritative, which is why both are reported.

> **Validity caveat.** The two runs differ and neither is a superset of the other. The primary run uses the standard predicted-class-confidence definition of ECE, but ran inference with `Resize` + `ToTensor` *without* the ImageNet normalisation used at training. That pre-processing skew collapses two of three detectors to a constant negative output (vehicle: 0 positive predictions on 3 600 frames, accuracy 0.250; traffic light: 81 positive predictions, accuracy 0.305), so their ECE of 0.74 and 0.60 is inflated by the skew, not by miscalibration alone. The cross-check run uses the correct normalised pipeline but scores the positive-class probability, which penalises the pedestrian model for being confidently negative. The observation is itself an important safety result, recorded as LS-8 and RR-5: a silent pre-processing mismatch is sufficient to disable an entire detector while every prediction still looks confident and well-formed.

Cost-optimal decision threshold. With $C_{\mathrm{FN}} = 100$ and $C_{\mathrm{FP}} = 1$ the two actions have equal expected loss at $\tau^* = C_{\mathrm{FP}}/(C_{\mathrm{FP}} + C_{\mathrm{FN}}) = 1/101 \approx 0.0099$, fifty times lower than the argmax rule. Evaluating $L = C_{\mathrm{FN}} \cdot \#\mathrm{FN} + C_{\mathrm{FP}} \cdot \#\mathrm{FP}$ on the pedestrian detector:

Pedestrian model	$\boldsymbol{\tau = 0.5}$	$\boldsymbol{\tau = \tau^*}$	**FP at** τ^*	**TN at** τ^*	**Recall at** τ^*
Uncalibrated ($T = 1.0$)	70 600	3 265	2 765	129	0.993
Temperature-scaled ($T = 2.9$)	70 600	**2 894**	2 894	0	1.000

The lowest loss is calibrated + τ^*, a 24× reduction from the default operating point — but nearly all of the gain comes from moving the *threshold*, not from calibrating (3 265 → 2 894 is 11 %, whereas $\tau = 0.5$ costs 70 600 either way). Both operating points are degenerate. At $\tau = 0.5$ the detector predicts “no pedestrian” on all 3 600 frames (recall 0.000), and calibration cannot rescue it because temperature scaling is monotone in the logit and never moves a decision taken at a fixed threshold. At τ^* it predicts “pedestrian” on all 3 600 frames (zero true negatives, accuracy 0.196): perfect recall bought with unconditional braking. **Neither point is deployable**, which is the quantitative core of the deployment recommendation.

Effect of T on the confidence gate $\theta = 0.6$. Because $T > 0$ preserves the sign of the logit, accuracy is identical at $T \in \{0.5, 1.0, 2.0\}$ (0.7967 in-ODD). What T changes is how often the constraint “if confidence $< \theta$, reduce speed to ≤ 15 km/h” fires: 3 138 frames at $T = 0.5$, 3 180 at $T = 1.0$, 3 228 at $T = 2.0$ (of 3 600). A small T sharpens the distribution towards 0 and 1 (Fig. 3) and fires the gate *least often*, so it is the **less safe** setting: it suppresses precisely the caution the constraint was written to produce. Conversely $T = 2.0$ fires on 90 % of frames, making the gate operationally useless. Accuracy can therefore never verify this constraint on its own; the additional property that must be measured is calibration of the probability itself, which is what ECE quantifies.

V-4: Out-of-Distribution Detection

Verifies: SC-5 **Hazard(s):** H-5, H-1 **Loss scenario(s):** LS-6
Constraint: An accompanying monitor reliably flags inputs that fall outside the ODD, preventing silent failure on conditions not covered by training data (fog, night).
Metric: AUROC separating the 3 600 in-ODD test frames from the 10 800 OOD frames (fog + night + unseen town); threshold ≥ 0.90.

Evidence (Anomaly Detection)

Model used	Detector	AUROC	Met?
Vehicle	MSP (max. softmax probability)	0.751	×
Vehicle	k-NN, $k = 5$, mean feature distance	0.983	✓
Pedestrian	not evaluated	—	not verified
Traffic light	not evaluated	—	not verified

Mean predicted-class confidence per condition (why MSP fails):

Model	In-ODD	Fog	Night	Unseen town
Pedestrian	0.945	0.976	0.976	0.958
Traffic light	0.963	0.814	1.000	—
Vehicle	0.904	0.609	0.862	—

Overall verdict: **≈ Partial**

The confidence table explains the result. A classifier's own softmax cannot be trusted as an OOD signal because nothing in the training objective ties confidence to familiarity: the pedestrian detector is *more* confident under fog and at night (0.976) than in-ODD (0.945), so an MSP monitor would flag OOD inputs *less* often than nominal ones for the most safety-critical detector. Aggregated over all OOD scenarios the vehicle model's MSP reaches AUROC 0.751, far below the 0.90 requirement.

The feature-based k-NN detector reaches AUROC 0.983, comfortably above threshold, because it measures distance in representation space instead of asking the classifier how sure it is: a shift that leaves the output layer confident still moves the embedding. Two caveats keep the verdict at *partial* rather than met. First, it was evaluated for the vehicle model only, so SC-5's "for all three detectors" is not discharged. Second, in this run the feature extractor was instantiated with random rather than trained weights and the in-distribution reference set was the test split itself rather than the training split. The high AUROC is therefore evidence that fog/night/town frames are separable by *some* feature-space statistic, not yet evidence that the deployed monitor generalises; re-running with the trained backbone and a training-split reference bank is required before SC-5 can be claimed.

V-5: Safe System Fallback

Verifies: SC-6, SC-7, SC-8, SC-9 **Hazard(s):** H-1–H-5 **Loss scenario(s):** LS-3, LS-6, LS-7, LS-8
Constraint: The system provides a safe fallback when a perception model fails or operates outside the ODD — even if a model produces incorrect output, the architecture limits propagation of the error to harm through fallback behaviour and human oversight.

Evidence (Design argument)

Fallback behaviour defined: *none is implemented in the system as specified.* The system description states explicitly that no safety mechanisms beyond the human operator are deployed. The following four-stage fallback is therefore specified as a *required design change*, not as existing evidence:

1. **Threshold change (SC-4).** Arm braking at $\tau^* = 1/101$ instead of 0.5, on temperature-scaled probabilities.
2. **Confidence gate (SC-6).** Where the probability lies in the ambiguous band around $\theta = 0.6$, limit speed to $\leq 15\,\mathrm{km/h}$ so that the braking distance stays inside the reliable detection range.
3. **OOD gate (SC-6).** On a k-NN monitor alarm, disable automated speed control, issue a takeover request and, if it is not answered within the takeover budget, execute a minimal-risk stop at the roadside.
4. **Independent stop authority (SC-8).** Derive stop decisions at signalised intersections from map data plus a state-aware signal classifier, never from the presence-only detector.

Bound on operator takeover. For the operator to bound H-1 the takeover must complete within $t_{\mathrm{TOR}} \leq (d_{\mathrm{detect}} - d_{\mathrm{brake}})/v$. At $v = 50\,\mathrm{km/h} = 13.9\,\mathrm{m/s}$ and $d_{\mathrm{brake}} \approx 12\,\mathrm{m}$ at $0.8\,g$, a usable detection range of $40\,\mathrm{m}$ leaves $t_{\mathrm{TOR}} \leq 2.0\,\mathrm{s}$. Nothing in the design supports that bound: the alert is visual only, shifts run for four hours, and takeover latency is never measured (LS-7).

System-level constraints addressed by design: SC-6, SC-7, SC-8, SC-9 are *specified* above but none is implemented or measured in the current system.

Overall verdict: × **Not met**

The decisive argument is structural. A fallback can only bound a hazard if its failure probability is both small and independent of the failure it is catching. Here neither holds: the operator shares the same single camera view of the world as the automation, receives no auditory alert, and — given a pedestrian recall of 0.191 — is not a fallback for the automation at all but its *primary* detector. Delegating 80 % of pedestrian detections to an unmeasured human channel is not a safety argument, and the constraint cannot be discharged.

5. Residual Risks and Mitigations

Residual Risks

ID	From	Residual risk (hazard left under-controlled)	Mitigation for the next version
RR-1	V-1	H-1 is essentially uncontrolled: the pedestrian detector misses 571 of 706 positive frames, so the automation contributes almost nothing to pedestrian protection. H-2 is partly uncontrolled (540 missed vehicles).	Class-balanced or focal loss; oversample pedestrian-positive frames; set the operating threshold from a recall target rather than 0.5; crop-and-zoom augmentation for distant pedestrians; move to a detector with localisation.
RR-2	V-2	H-1 and H-2 remain open under adversarial input. The FGSM recall drop is unknown, and a 10×10 pixel trigger achieves ASR = 1.00 on a poisoned model.	Run the provided FGSM harness as an ε-budget acceptance gate; PGD adversarial fine-tuning; activation-clustering screening of the training set; provenance control and hashing of every training batch.
RR-3	V-3	H-1 and H-4 remain open through LS-5: confidence carries no usable information, so no confidence-gated fallback can be trusted. The cost-optimal threshold restores recall only by braking on 80 % of frames.	Calibrate on a held-out split rather than test; add ensembles or MC dropout for epistemic uncertainty; report per-condition ECE; re-tune the cost matrix against an achievable false-positive rate.
RR-4	V-4	H-5 remains partly open: no OOD monitor is verified for the pedestrian or traffic-light detector, and the verified k-NN result rests on random features and a test-split reference bank.	Rebuild the k-NN/Mahalanobis bank per model from trained backbone features on the *training* split; report per-scenario AUROC and FPR95; add an image-quality monitor.
RR-5	V-3 note	LS-8 is unmitigated: a silent pre-processing mismatch disables a detector completely while every output still looks confident.	Serialise the pre-processing with the weights; assert an equivalence test against stored reference activations at start-up (SC-9); monitor the output-rate distribution for plausibility.
RR-6	ODD	37 % of 3-way ODD combinations are untested; no frame combines two adverse conditions and rain is absent entirely, so no verdict extends there.	Collect combinatorial CARLA splits (fog × night, night × town, rain, dusk) until 3-projection coverage reaches 0.90.
RR-7	V-5	Every hazard H-1–H-5 ultimately relies on an operator whose takeover latency is unmeasured, unalerted and subject to a four-hour vigilance decrement.	Implement the four-stage fallback of V-5; add auditory takeover alerts; cap shifts and add driver-monitoring; measure takeover latency per operator and treat the 95th percentile as the budget in the t_{TOR} bound.

Known design limitations

These follow from the system as specified; no model metric in this report can close them.

ID	Limitation	Hazard left open
DL-1	The traffic-light detector reports presence only, not state, so even at recall 0.991 its high score contributes nothing to intersection safety.	H-3
DL-2	No sensor redundancy: a single camera is the only perception input, so every perception fault is a single point of failure with nothing to cross-check it.	H-1–H-5
DL-3	The human operator is the only fallback, is alerted visually only, and works 4-hour shifts; takeover latency is never measured.	H-1–H-5
DL-4	All training and test data are CARLA renderings, so no claim transfers to real camera imagery (sensor noise, motion blur, real material appearance).	H-1–H-5
DL-5	Frames are resized to 224×224, destroying the pixel support of distant pedestrians — exactly where early braking matters most.	H-1
DL-6	Binary whole-image classification gives no localisation, so the planner cannot tell whether a detected pedestrian is in the ego path or on a sidewalk; the assumed-perfect distance oracle hides this gap.	H-1, H-4

6. Deployment Recommendation

Recommendation: ☐ Deploy ☐ Deploy with restrictions ☒ **Do not deploy**

Conditions / restrictions. No public-road operation, with or without a safety operator; only closed-course or simulation-in-the-loop testing with no uninvolved road users present. Re-assessment requires, as a minimum: pedestrian recall ≥ 0.90 and vehicle recall ≥ 0.85 at a declared threshold (SC-1, SC-2); FGSM recall drop < 10 pp and ASR $\leq 1\,\%$ (SC-3); post-calibration ECE ≤ 0.05 on a held-out split (SC-4); AUROC ≥ 0.90 for *all three* detectors from trained features and a training-split bank (SC-5); the four-stage fallback implemented with measured takeover latency (SC-6, SC-7); and 3-projection ODD coverage ≥ 0.90.

Justification. Four of five verifications fail and the fifth is only partial. V-1 is disqualifying on its own: pedestrian recall is 0.191 against a 0.90 requirement, missing 81 % of the frames containing the road user whose injury is loss L-1, so H-1 is effectively uncontrolled by the automation. V-2 shows the constraint is not merely unverified but violated, a 10×10 pixel trigger flipping every pedestrian-positive frame (ASR $= 1.00$) while clean-set metrics stay plausible. V-3 shows all three detectors overconfident beyond the ECE limit before and after temperature scaling, so no confidence-gated fallback can be trusted, and the cost-optimal threshold that restores recall to 1.000 does so by braking on *every* frame. V-4 is only partial, with MSP at 0.751 and the pedestrian model growing *more* confident on out-of-ODD input. V-5 cannot be discharged at all, because the only fallback is a human who shares the same single camera view, gets no auditory alert, and is in practice the primary pedestrian detector rather than a backup.

These failures compound rather than offset: an under-performing detector (V-1) whose errors are indistinguishable from correct outputs (V-3), on inputs whose novelty is not reliably flagged (V-4), with no independent channel to catch the error (V-5). No restricted envelope within the declared ODD can be defended on this evidence.

Statement of Authorship

I hereby declare that I have independently written the present work and have not used any sources or aids other than those stated.

Texts, trains of thought, concepts, graphics, etc. from external sources that I have directly or indirectly adopted in my work I have clearly marked as such and provided with complete references to the respective source. All other content of this work (text parts, figures, tables, etc.) without corresponding references originates in the copyright sense from me.

I have exclusively used AI systems whose use was approved by the examiner as an aid.

Excluded from the requirement of identification are orthographic or grammatical corrections, translations, and improvements to formulations that do not alter meaning.

I am aware that AI-generated texts offer no guarantee for the quality of content and text. I therefore declare that I have used AI systems only as an aid, that I have listed and marked the AI-generated content, critically reviewed it, and that my independent contribution predominates in this work.

I confirm that I have fully understood the content of my work and can represent it independently.

Place, Date

Signature

A. Additional Material

Reproduction map

Result in this report	Produced by (run all cells / execute the script)
Architecture, training curves, class balance (Section 1.3, Table 1)	`Excercise 3/3 Fundamentals.ipynb`
Recall / precision / F1, confusion matrices (V-1)	`Excercise 4/4 Model Testing and Validation.ipynb`
k-projection ODD coverage (Section 2)	`odd_coverage/k_projection_coverage.py`
Grad-CAM overlays (V-1, Fig. 2, Fig. 4)	`explainability/explain_conditions.py`, `explain_correct.py`, `explain_errors.py`
Backdoor clean recall and ASR (V-2)	`Excercise 5/backdoor_attack.py`
FGSM recall drop (V-2, *pending*)	`adversarial/fgsm_robustness.py` (add `--limit 100` for the 100-image subset)
Accuracy and confidence-gate counts vs. T (V-3, Table 3)	`Excercise 5/temperature_scaling.py`
ECE, reliability diagrams, τ^* loss table (V-3, primary run)	`Excercise 9/9 Uncertainty Quantification.ipynb`
ECE cross-check on the normalised pipeline (V-3, Table 2)	`Excercise 8/8 Adversarial ML.ipynb`
MSP and k-NN AUROC, mean confidences (V-4)	`Excercise 7/7 Anomaly Detection.ipynb`

Training and validation loss per epoch

	Pedestrian		Traffic light	
Epoch	**Train**	**Validation**	**Train**	**Validation**
1	0.5128	0.5690	0.1975	0.1126
2	0.4310	0.5949	0.0826	0.1160
3	0.3676	0.5181	0.0712	0.1121
4	0.3174	0.6950	0.0527	0.0730
5	0.2686	0.7138	0.0383	0.0770

Table 1: Binary cross-entropy per epoch. The pedestrian detector's validation loss rises from epoch 3 onward while its training loss keeps falling — overfitting on the minority class, consistent with the recall failure in V-1. The vehicle detector's curves were not retained in the notebook output.

Calibration cross-check on the normalised pipeline

Model	ECE (single)	T^*	ECE (temp. scaled)	Met?
Pedestrian	0.749	3.0	0.609	×
Traffic light	0.245	1.9	0.207	×
Vehicle	0.226	1.5	0.220	×

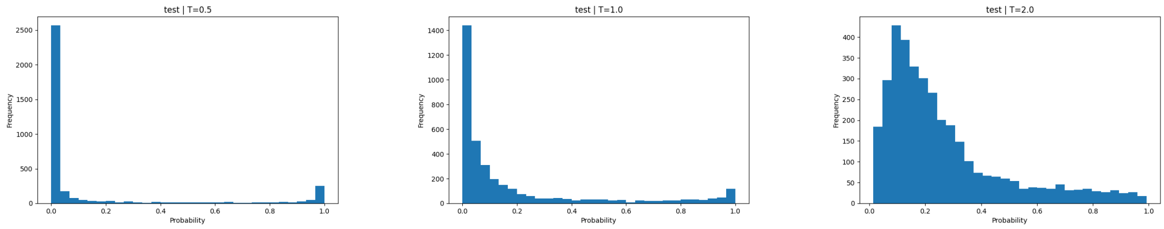
Table 2: V-3 cross-check. Inference run with the ImageNet normalisation used at training, ECE computed on the positive-class probability with T chosen by grid search. Every value still exceeds the 0.05 limit, so the V-3 verdict does not depend on which run is taken as authoritative. The pedestrian figure is high because this definition penalises a model that is confidently *negative* on 80 % of frames.

Per-condition accuracy and confidence-gate counts

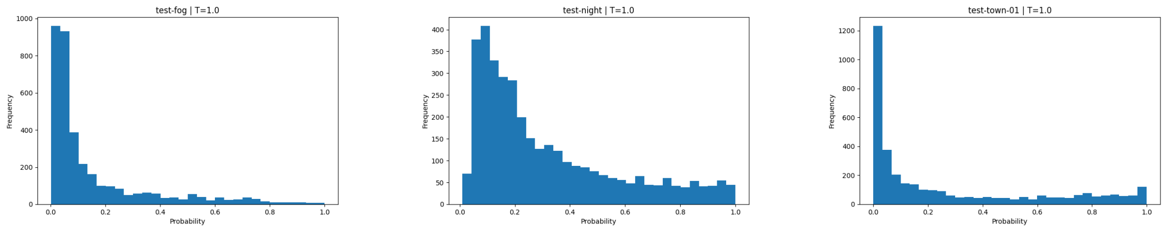
Split	T	**Accuracy**	**Frames with** $p < 0.6$ **(of 3 600)**
`test` (in-ODD)	0.5	0.7967	3 138
	1.0	0.7967	3 180
	2.0	0.7967	3 228
`test-fog`	0.5	0.7636	3 340
	1.0	0.7636	3 376
	2.0	0.7636	3 454
`test-night`	0.5	0.7106	2 926
	1.0	0.7106	3 017
	2.0	0.7106	3 169
`test-town-01`	0.5	0.7517	2 781
	1.0	0.7517	2 840
	2.0	0.7517	2 983

Table 3: Exercise-5 pedestrian detector. Accuracy is invariant in T because dividing a logit by $T > 0$ cannot change its sign, so the decision at threshold 0.5 is unchanged; only the confidence-gate count moves.

Predicted-probability distributions



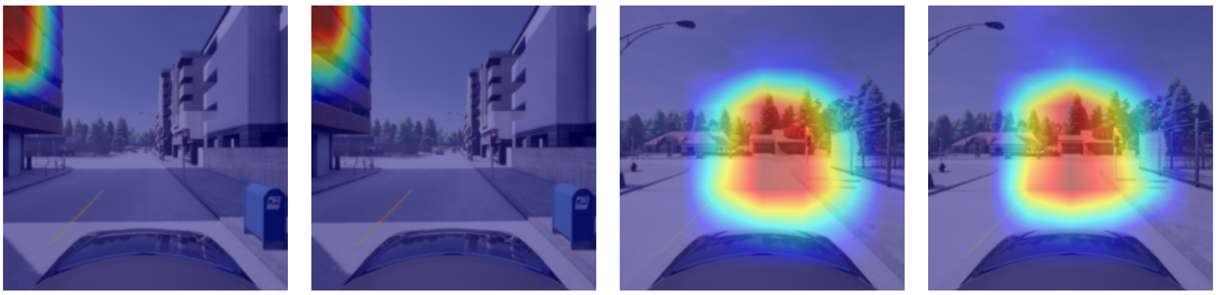
(a) in-ODD, $T = 0.5$ (b) in-ODD, $T = 1.0$ (c) in-ODD, $T = 2.0$



(d) fog, $T = 1.0$ (e) night, $T = 1.0$ (f) unseen town, $T = 1.0$

Figure 3: Distribution of the predicted pedestrian probability p_T. Raising T pulls mass away from the extremes towards the centre; lowering it sharpens the distribution towards 0 and 1 and therefore fires the $\theta = 0.6$ confidence gate least often. At $T = 1.0$ the number of frames *above* 0.6 is 420 in-ODD but 583 at night and 760 in the unseen town (Table 3), i.e. the detector is not less confident out of domain — only under fog (224) does it become more cautious.

Grad-CAM on correct and misclassified in-ODD frames



(a) correct (b) correct (c) misclassified (d) misclassified

Figure 4: Grad-CAM on the in-ODD test split. On correctly classified frames the attribution sits on a building facade in the upper-left corner, not on any road user — a spurious correlation with the training town. On misclassified frames it forms one large blob centred on a vehicle in the middle of the road, i.e. the pedestrian decision is driven by vehicle-shaped structure. The frames within each pair are temporally adjacent, because the selection scripts take the first matching frames of the split rather than a random sample; they are therefore not independent examples.